Lab 1 — Use-Case Flow Document

Problem Statement #26 — Peer-to-Peer Carpooling & Expense Splitter

Use Case	UC-03: Book Ride (Match & Reserve Seat)
Primary Actor	Rider Commuter
Secondary Actors	Driver Host, Payment Gateway
Preconditions	Rider Commuter is logged in; at least one Driver Host trip listing exists that satisfies the 1.5 km detour tolerance.
Postconditions	Seat is reserved, fuel cost split is finalized, and payment is confirmed for all parties.

Main Success Scenario

1. Rider Commuter logs into the app and enters origin, destination, and desired travel time.

2. System (Search & Match Ride) displays Driver Host trip listings that fall within the 1.5 km detour tolerance and time window.

3. System calculates and displays the estimated fuel cost share for each matching trip (<<include>> Calculate Fuel Cost Split).

4. Rider Commuter selects a trip and taps "Request to Book."

5. System sends a booking request notification to the Driver Host.

6. Driver Host reviews and accepts the request.

7. System confirms the seat reservation and finalizes the fuel cost split among all confirmed passengers.

8. System prompts Rider Commuter to pay; Rider Commuter authorizes payment via the Payment Gateway.

9. System sends a booking confirmation with trip details and cost breakdown to both Rider Commuter and Driver Host.

10. Use case ends successfully.

Alternate Flows

6a. Driver Host Declines the Request

- 6a1. System notifies the Rider Commuter that the request was declined.
- 6a2. System displays other matching trips within the detour tolerance.
- 6a3. If no other trips are available, Rider Commuter is prompted to widen the search radius or choose a different departure time.

8a. Payment Declined

- 8a1. System displays a "payment failed" message to the Rider Commuter.
- 8a2. Rider Commuter is prompted to try another payment method.
- 8a3. If payment fails again after 2 attempts, the booking is cancelled, the seat is released back to the Driver Host's listing, and both parties are notified.